TrueSite

BIM ON SITE · AS-BUILT AI

# See the design where you build. Check the build against it.

AR × BIM × AI for construction. Scan a QR code on site and the BIM model appears at true scale in the phone's browser. Tap **Check as-built**: Gemini compares what is built with the design, pins the differences and drafts the issue.




**GOOGLE CLOUD AI BUILDER CUP 2026 · JAPAC**

Theme: Future of Work & Enterprise Productivity · Team: Wai Lin, Aung Nyein Chan (Michael)

THE PROBLEM

# Crews build without the model in front of them

**14+ h**

a week per project team member go to hunting for project data, resolving conflicts and rework

FMI & PlanGrid, 2018

**48%**

of all rework on US jobsites comes from miscommunication and poor project data

FMI & PlanGrid, 2018

**1%**

a year: construction's labour-productivity growth over two decades, against 2.8% for the world economy

McKinsey Global Institute, 2017

The model exists: Singapore requires BIM submissions for buildings over 5,000 $m^2$ (BCA). It just stays in the office, while decisions get made on site and mistakes surface weeks later as rework.

Sources: FMI & PlanGrid, "Construction Disconnected" (2018, nearly 600 construction leaders); McKinsey Global Institute, "Reinventing construction" (2017); Building and Construction Authority, Singapore

THEME · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

*“Transform how individuals and organizations access knowledge, collaborate, make decisions, and complete everyday work.”*

### Access knowledge

The BIM model at true scale, where the work happens, from a printed QR code and a phone.

### Make decisions

Gemini checks the build against the design and pins what is off, with why it matters.

### Complete everyday work

Issues drafted in the crew’s language, questions answered from the model, a site report written.

**TrueSite:** point a phone at a QR code and the BIM model appears over the real site at true scale. Tap **Check as-built**: Gemini compares what is built with the design, pins the deviations and drafts the issue.

Users: field engineers, foremen, site architects · Buyers: contractors

WHY NOW · THE FUTURE OF CONSTRUCTION

# BIM is becoming the rule. AI makes it work on site.

### BIM is required

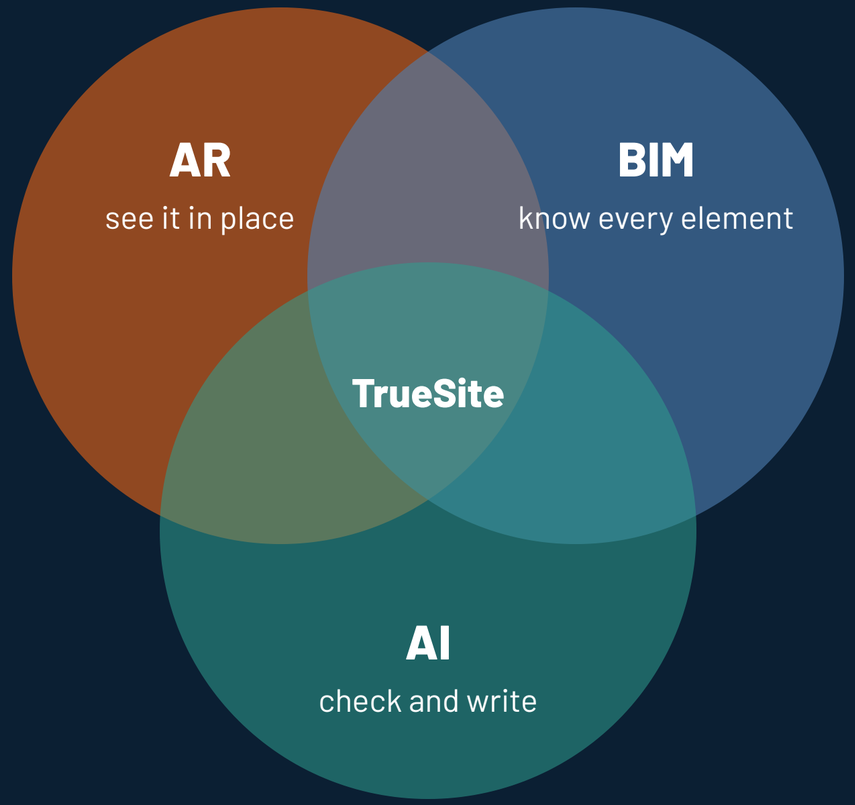
Singapore requires BIM submissions for buildings over 5,000 $m^2$. The UK has required BIM on centrally procured public projects since April 2016.

### Construction is behind

One of the least digitised industries: second to last in the US and last in Europe on McKinsey's digitisation index.

### The next step

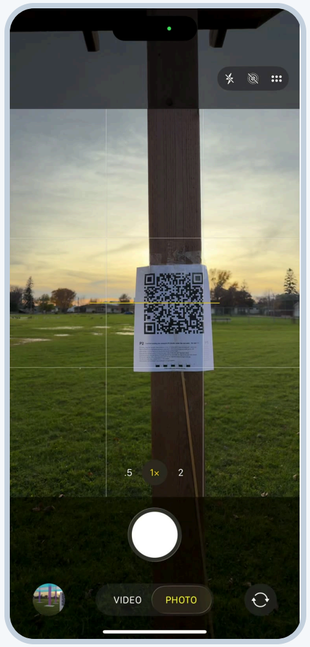
Take the model to the crew, and let AI read it: check the build against the design, answer questions, write the issues.




Sources: Building and Construction Authority, Singapore; UK Government BIM mandate (4 April 2016); McKinsey Global Institute, "Reinventing construction" (2017)

HOW IT WORKS

# A link and a QR code. Nothing to install.



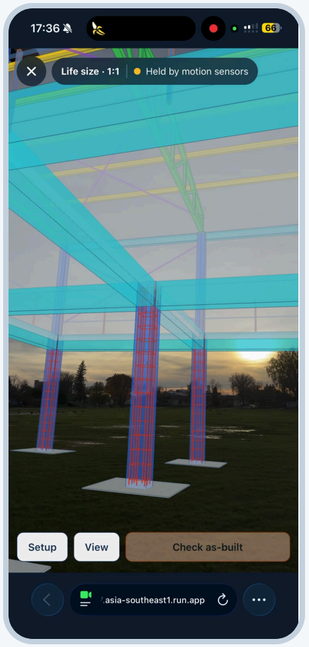

## 1 · Scan

The camera reads the code; TrueSite opens in the browser



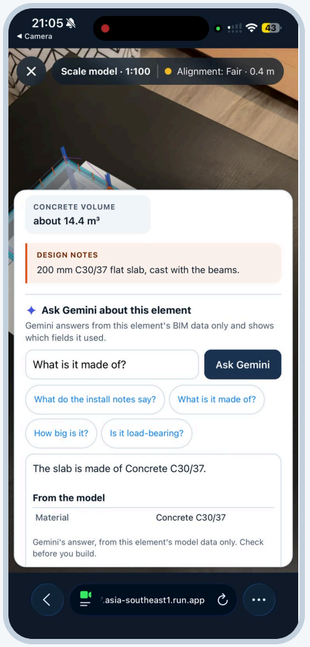

## 2 · Lock

Tilt for 10 s to calibrate; the code's corners give the pose




## 3 · See

The design at 1:1, held by the motion sensors



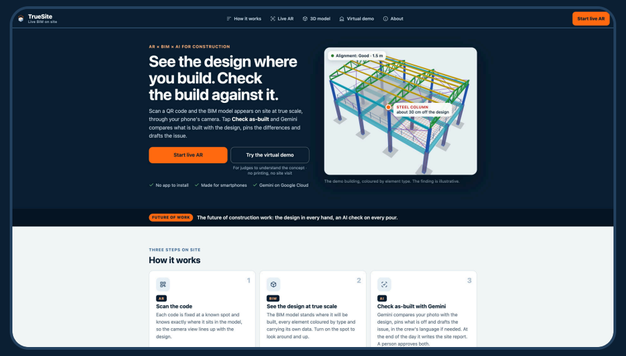

## 4 · Tap, ask, check

Any element's BIM data; Gemini answers and checks

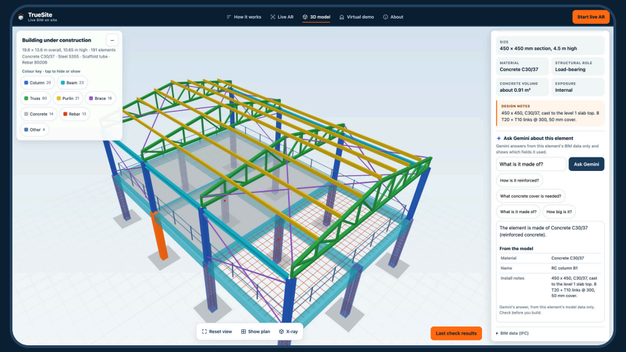
Full screens from our iPhone field test, 17 Oct 2026, on the deployed Cloud Run service

PRODUCT TOUR

# One prototype, five ways in



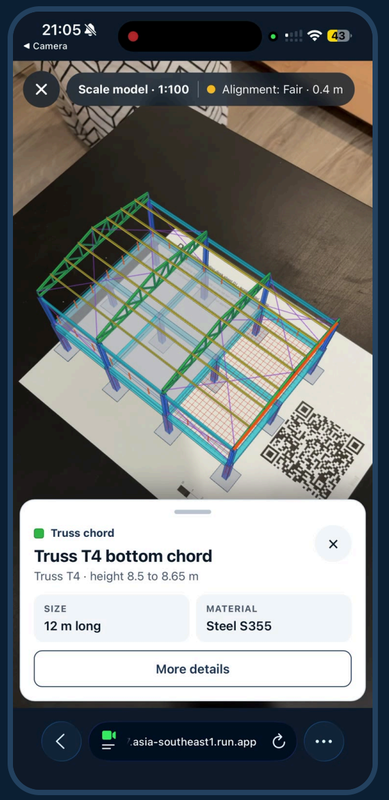

**Start page:** try, print, watch




**3D model viewer** with Ask Gemini



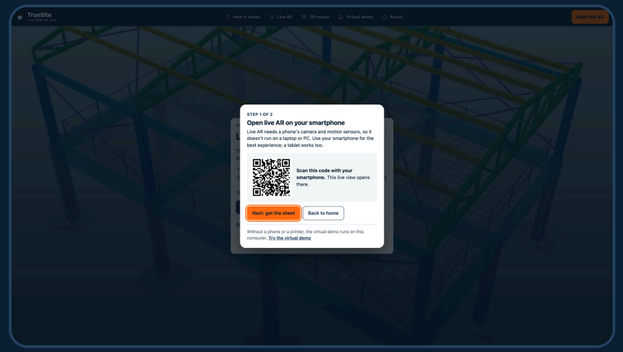

**Live AR** on site, 1:1




**Desk sheet** at 1:100



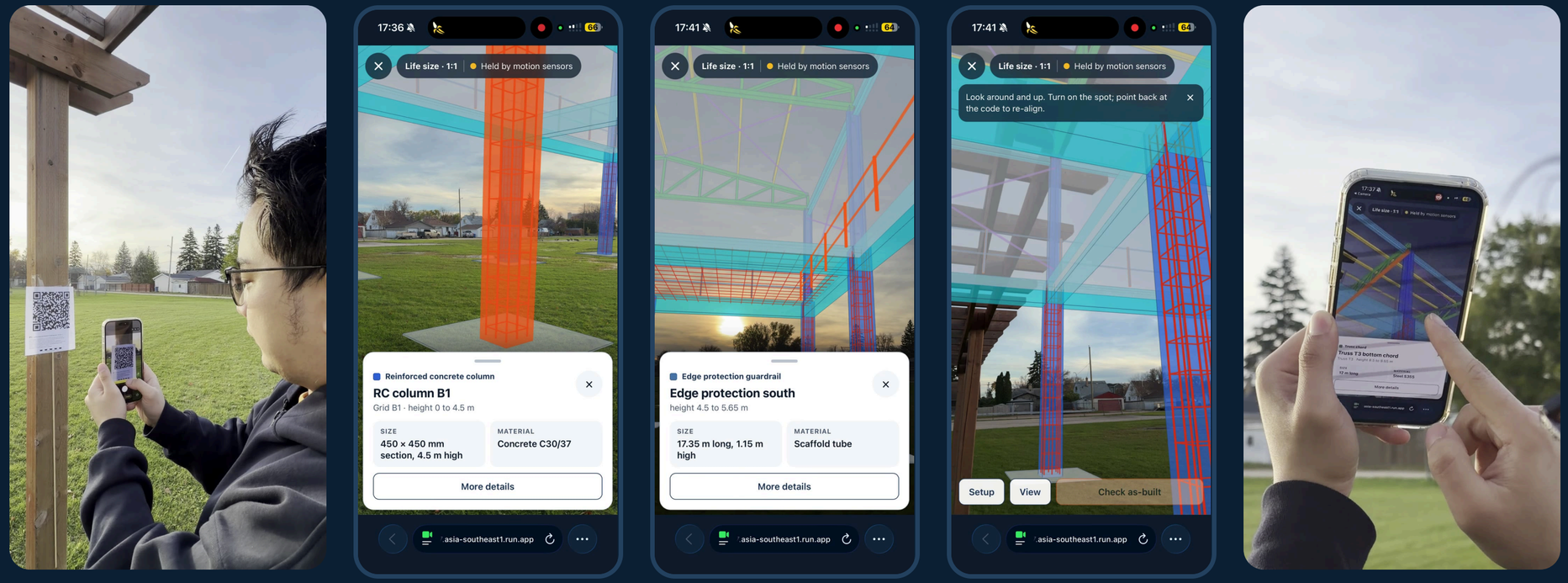

**Virtual demo,** live Gemini check




**Laptop:** handed to a phone by QR

LIVE AR · FIELD TEST, 17 OCT 2026

# Life size on a real field, from an iPhone’s browser




The site sheets on posts put the building at 1:1. It held still while turning and looking up, and every element answered a tap with its BIM data, including rebar cages and edge protection.

Real footage on the deployed Cloud Run URL · no app, no headset · iPad steadiness fixed on 18 Oct, iPhone path unchanged

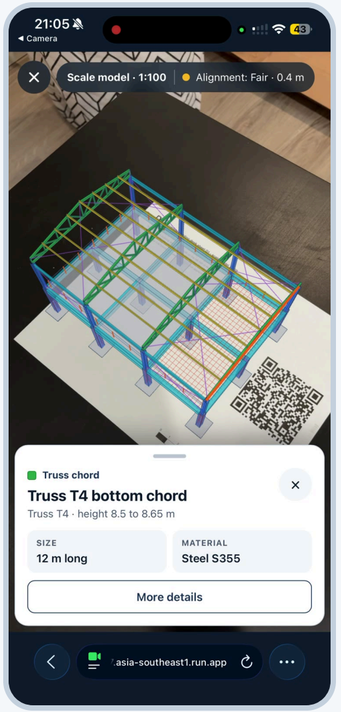
LIVE AR · DESK MODE AND VIEW CONTROLS

# The whole building on a desk, at 1:100

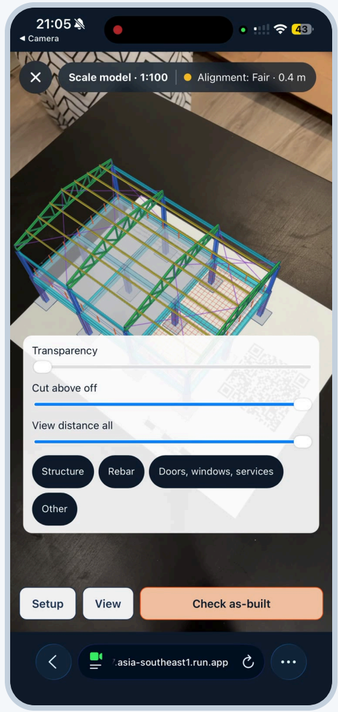
Print the desk sheet and scan it: the building stands on the page as a 1:100 model. Walk around it and tap any element.

## View controls, live and on site

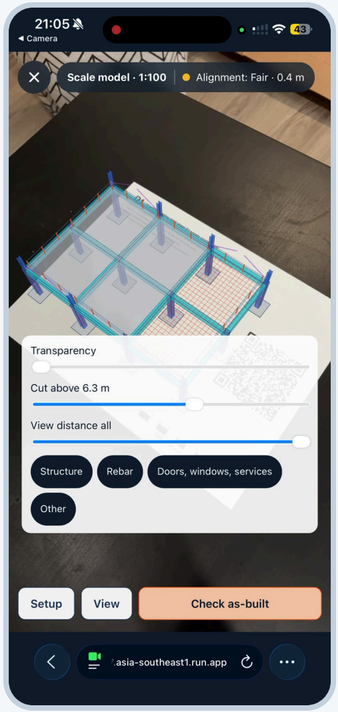
- Transparency
- Cut above: a section at any height
- View distance
- Layers: structure, rebar, doors and windows, services
- Elements coloured by type



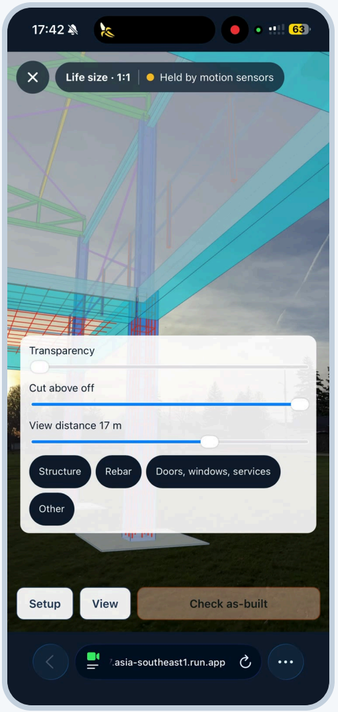

Tap: truss chord




The View panel




Cut above 6.3 m

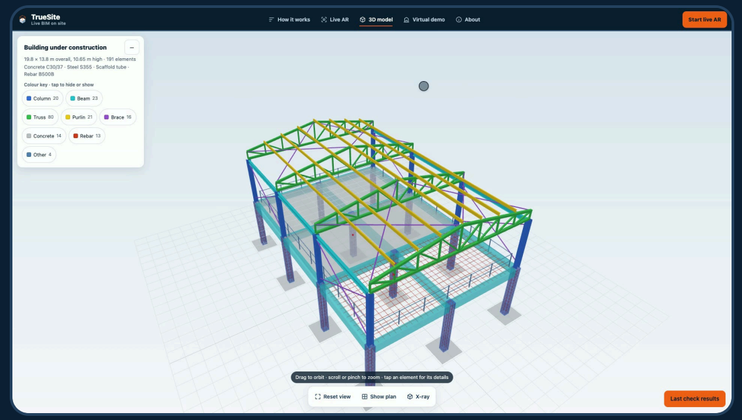



Same controls at 1:1

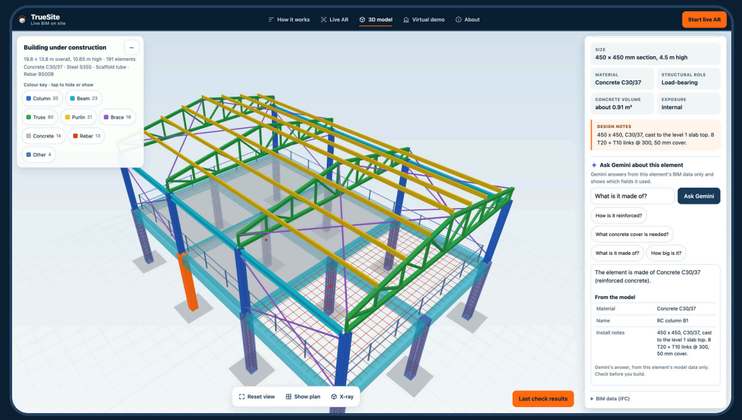
Full screens from the iPhone · the desk sheet is a US Letter page with a 62 mm code

BUILT-IN 3D MODEL VIEWER

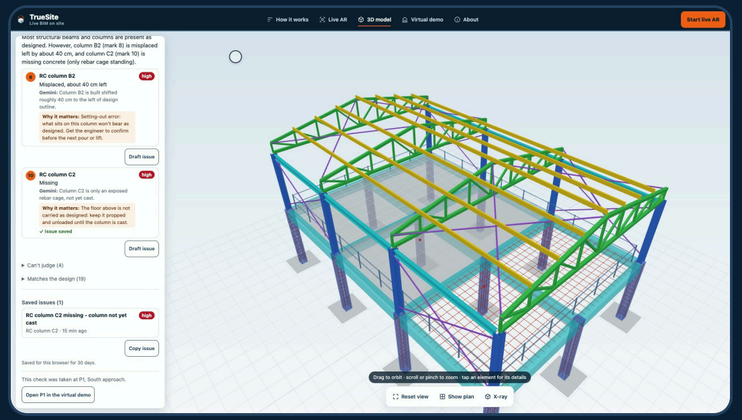
# The full BIM model, in the office or on the go




**Orbit the whole building,** coloured by type, with a plan view




**Tap an element:** its BIM data, and Ask Gemini about it




**The last check's findings** and saved issues, pinned on the model

One model for the whole team: what the crew sees on site, the engineer sees in the office, with the same findings, saved issues and Gemini answers. The 3D model and the virtual demo link to each other.

Screens from the deployed app (laptop) · the same viewer works on phones and tablets

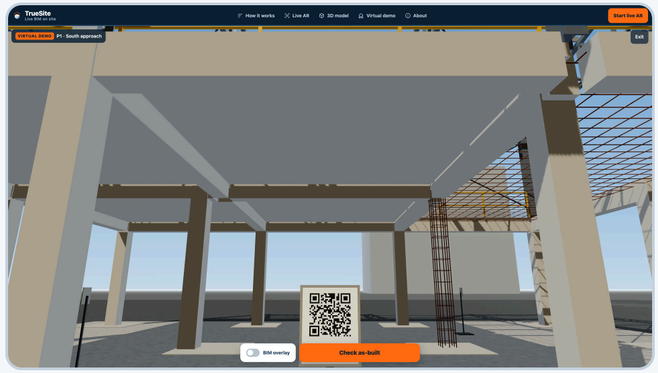
VIRTUAL DEMO · BUILT FOR JUDGES

# No building yet, so we built a whole 3D site to test on

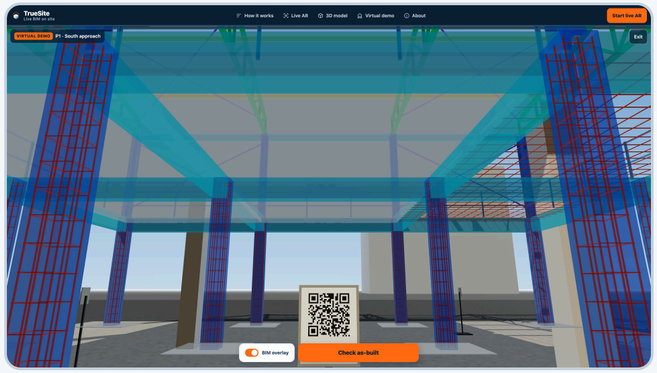
Our demo building doesn't exist yet, so live AR has nothing real to check. We built a complete 3D virtual environment: a simulated as-built site with **ten defects planted on purpose**, from a shifted column to missing edge protection.

1. Step in at a QR spot, P1 to P4
2. Look around with the BIM overlay on or off
3. Tap **Check as-built**: live Gemini judges the view
4. A scorecard shows what it caught and missed

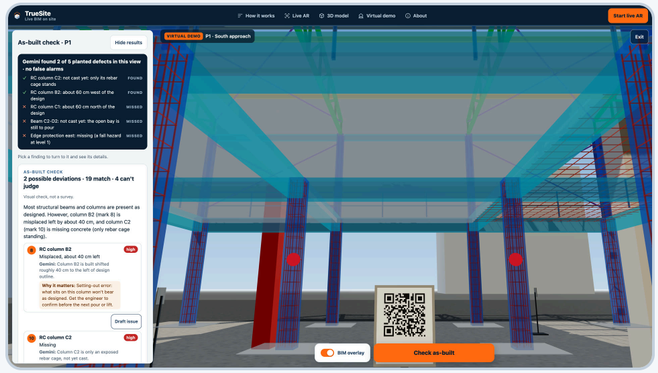
Gemini never sees the planted list. Runs on any device, laptops included.



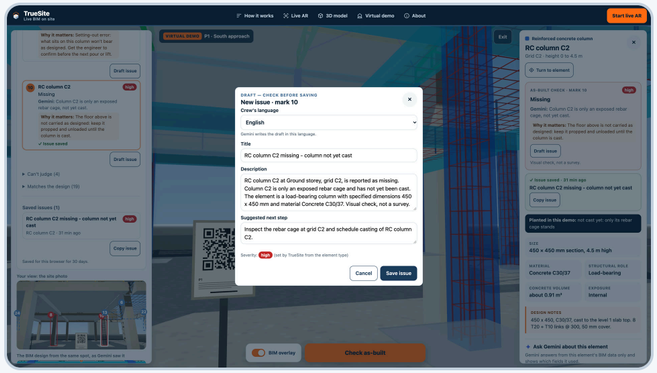

The simulated as-built site




BIM overlay on




Scorecard and findings




The issue Gemini drafted

Live at truesite-1067044308557.asia-southeast1.run.app/#virtual · simulated views, labelled as such

THE GEMINI AS-BUILT CHECK

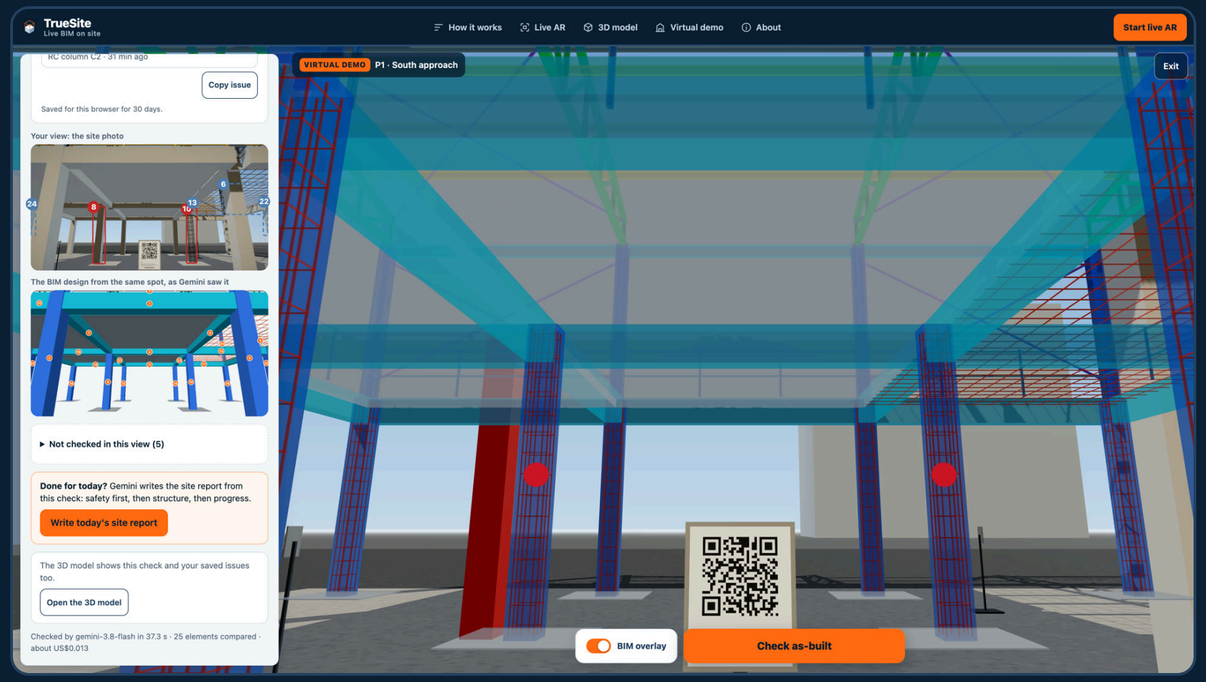
# Gemini judges. Code decides how serious.

**1 · Three images**
The photo, the design rendered from the same pose with numbered marks, and the photo with each element’s outline

**2 · Gemini 3.8 Flash**
Judges each mark at its outline; returns structured JSON

**3 · Server rules**
One finding per mark, offsets in tens of cm, low confidence becomes “Can’t judge”; severity from the element class

**4 · Pinned, explained, drafted**
Pins on the model, “Why it matters” from fixed rules, the issue drafted for a person to save

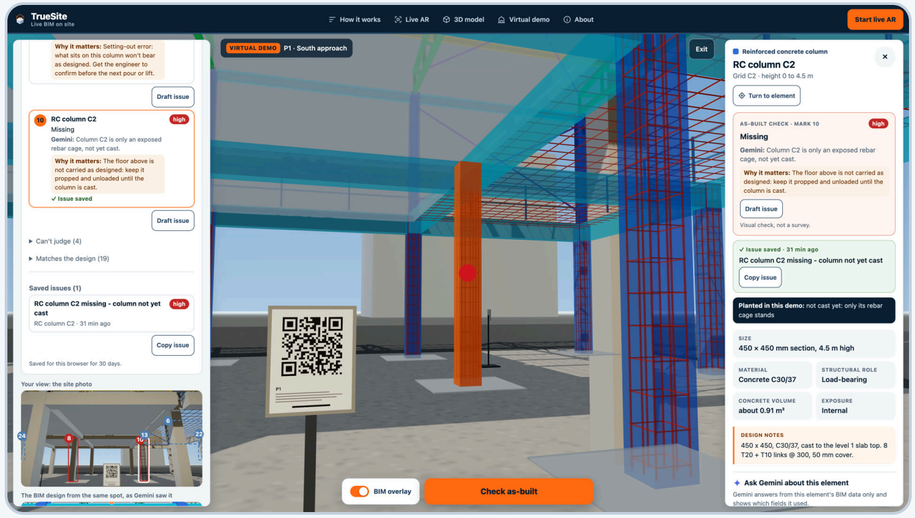



A live check in the virtual demo: the photo next to the design render Gemini compared. Gemini found 2 of the 5 planted defects in this view, with no false alarms.

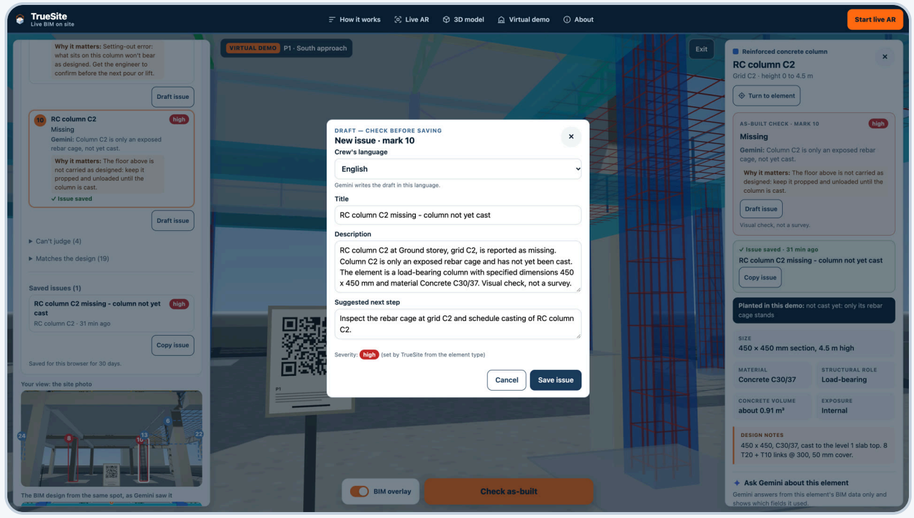
Every finding is labelled “Visual check, not a survey.” Gemini has no tools and no write access; its output is schema-checked.

ASK · DRAFT · SAVE

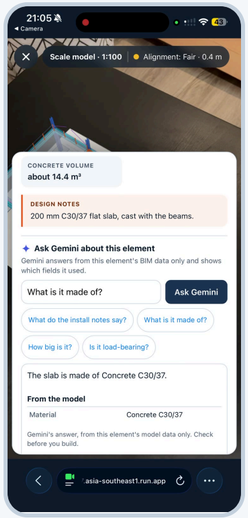
# Gemini drafts. People decide.




**A finding** with "Why it matters" from fixed rules, and the saved issue




**Draft issue** in one of eight crew languages; a person edits and saves it




**Ask** on site

Answers come only from the element's own BIM properties, with the fields shown. Gemini also writes the day's site report from the checks, safety first. Nothing Gemini writes is saved without a person.

Copy issue puts a saved issue on the clipboard for any tracker

INTEGRATION · OPEN BIM

# Works with the BIM you already have

1 · IN

### Any BIM tool's IFC

Revit, Archicad, Tekla, Allplan, Vectorworks and most BIM tools export IFC, the open ISO 16739 standard

2 · PREPARED ONCE

### TrueSite pipeline

- glTF model, every element keyed by its IFC GlobalId
- Element data: type, storey, size, material, properties, quantities
- QR sheets and a set-out plan

3 · USED ANYWHERE

### Any phone or laptop

Live AR, the 3D model and the virtual demo in the browser; Gemini on Google Cloud reads the same element data

4 · OUT

### Back to the team

Issues copied into any tracker, and the day's site report. Next: BCF export and connectors for common data environments

No vendor lock-in: the model stays in the open IFC format, and the crew needs nothing but a browser.

Built on the open-source IfcOpenShell, which reads IFC2x3, IFC4 and IFC4x3 · tested so far with our own IFC4 model

SOLUTION ARCHITECTURE · GOOGLE CLOUD

# The phone does the vision. Google Cloud does the judging.

### Phone browser

React + three.js, one web app

- QR reading in a zxing-wasm worker
- Pose from the code's corners; 10 s self-calibration
- Sensor fusion, gravity levelling
- Overlay at true scale
- ID buffer → numbered marks, design render, outlines

**3 images** →

### Cloud Run

asia-southeast1 · FastAPI and the web app on one origin

Limits and caps · schema repair · severity by rule · sessions, no accounts

→

### Gemini 3.8 Flash

Agent Platform, global · structured JSON · busy or hung: 3.5 Flash, then Flash-Lite

### Firestore

Check runs · issues · atomic daily cap

### Cloud Storage

Photos · renders · 30-day lifecycle

**Also:** Cloud Build and Artifact Registry for source deploys · Secret Manager (API-key fallback only) · budget alerts · a least-privilege service account

A check costs about US$0.015 in Gemini tokens, measured on the deployed service

ROBUSTNESS · MEASURED, NOT CLAIMED

# What we measured

| Measurement | Result |
|---|---|
| Pose accuracy · simulation, 0.5 px corner noise, 0.5–3 m | median 3.6 mm, 0.66° |
| Self-calibration · simulation, 30 tilted views | focal length within 1% |
| Steadiness on a stand · ray-traced test videos | 0.02–0.2 px jitter a frame (was 1–44 px) |
| As-built check · simulated site, 16 planted defects in view, 3 live runs | found 12, 11, 7 · false alarms 5, 8, 0 |
| Cost of one check · deployed service | about US$0.015 |
| Automated tests | 549 web · 527 server · 88 BIM · 168 end-to-end |

End to end on a laptop, an iPad and an iPhone profile, plus ray-traced camera videos with exact ground truth for live AR. Mutation, property and API fuzz testing on the core logic: 4,162 generated API cases, no server errors.

Simulation numbers are not field accuracy; the overlay error on a device at 1–3 m is not measured yet. We publish what we measure.

INNOVATION · PRIOR ART, OWNED

# AR and BIM exist. TrueSite adds the AI.

ON THE MARKET

## AR + BIM

Dalux · Trimble · GAMMA AR · XYZ Reality

They place the model with markers or survey hardware and show it on site. Most need an installed app or dedicated hardware.

TRUESITE

## AR × BIM × AI, in one link

**AR**
The model at true scale, from a QR code and a browser

**BIM**
Every element's own data, a tap away

**AI**
Gemini checks the build and writes the issues

The new idea: the QR code fixes the camera's exact pose, so we render the design from that same viewpoint, mark every visible element, and let Gemini judge each one against its own outline.

Built from scratch during the hackathon (repository started 11 Oct 2026, every commit dated)

BUSINESS CASE

# Free to view. Priced per field seat. Pennies to run.

### Who pays

Viewing is free for anyone who scans a code. Contractors pay per field seat for checks, Ask and issues: US$19 a month (proposed, to validate in the pilot).

### Cost to serve

About US$0.015 a check, measured. Even a heavy user at 20 checks a day for 22 days spends about US$6.60 of Gemini a month.

### Singapore first

S$47–53 billion of construction demand projected for 2026, and BIM submission already mandatory over 5,000 $m^2$. Then Taiwan and the rest of JAPAC.

### An 8-week pilot

One or two Singapore contractors, 10 field seats. Success: deviations caught before pouring, time to answer a model question on site, false-alarm rate.

Sources: Building and Construction Authority, Singapore (Jan 2026 outlook; BIM e-submission); check cost logged on our deployed service

IMPACT

# Less rework. Faster sign-off. Safer sites.

### Catch it before it’s poured

A missing cage or a shifted column found on site, while it can still be fixed, not weeks later as rework.

### Safer for workers

Guardrail and edge-protection deviations are always rated high, and issues reach the crew in their own language.

### Anyone on the crew

A phone and a printed code: no app, no headset. Subcontractors join by scanning.

**Next**

A real-site pilot with labelled photos to measure the check · world-locked AR on Android and walk-inside on iPhone (built behind a flag, not yet device-tested) · issue export to trackers · multi-code site set-up

Community benefit: safer sites and fewer wasted materials, for crews of every language

TEAM



## Wai Lin

Product and code: live AR, pose and sensor fusion, the Gemini pipeline, Google Cloud



## Aung Nyein Chan (Michael)

Architect: the BIM model, field validation and visuals

**Try it:** truesite-1067044308557.asia-southeast1.run.app

**Watch:** the demo video on the start page

**Read the code:** github.com/wl-geek/truesite

# Thank you.

TrueSite

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